

9708/m/j/2024/v32

1-C:

Formula of MU= **Change in TU/Change in Qty**

| quantity consumed / bottles | total utility | MU |
|-----------------------------|---------------|----|
| 0                           | 0             | 0  |
| 1                           | 30            | 30 |
| 2                           | 40            | 10 |
| 3                           | 48            | 8  |
| 4                           | 54            | 6  |
| 5                           | 58            | 4  |

The question presents total utility data from consuming water bottles. Marginal utility (MU) is calculated as the change in total utility from consuming one additional unit. Between the 2nd and 3rd unit, total utility rises from 40 to 48, indicating a marginal utility of 8. Therefore, statement C is correct.

**Why others are incorrect:**

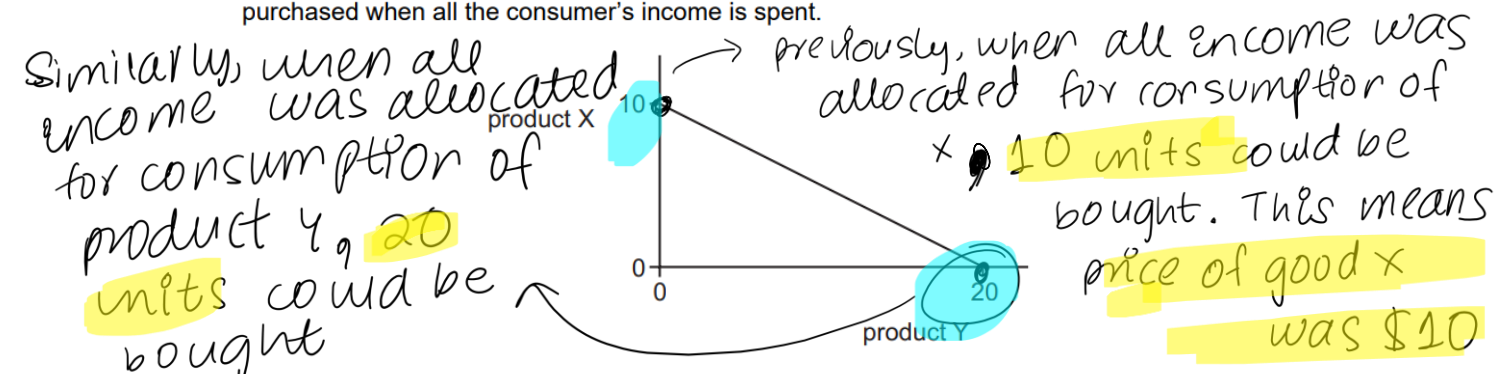
- **A** is incorrect because marginal utility is **declining**, not increasing, which is consistent with the **law of diminishing marginal utility**.
- **B** cannot be concluded from the data; consumers can generally switch to consumption of another good to increase their total utility.
- **D** is wrong because MU for the 5th unit is  $58 - 54 = 4$ , not 2.

2-C:

The image below explains why key C will be correct

2 A consumer has \$100 to spend on two products, X and Y.

The budget line shows the different possible combinations of products X and Y that can be purchased when all the consumer's income is spent.



If the price of product Y increases to \$10, what will be the maximum number of units of product X and product Y that the consumer can now purchase?

|   | product X | product Y |
|---|-----------|-----------|
| A | 5         | 10        |
| B | 5         | 20        |
| C | 10        | 10        |
| D | 20        | 10        |

price of good Y was \$5

Income = 100

New price of Y = 10

Qty purchased =  $\frac{100}{10} = 10$  units of Y

same units of good X as before can be purchased i.e. 10 units

3-A

### Net Social Costs = SC - SB

- For Net Social Costs to increase, either Social Costs should increase or Social Benefits should decrease.
- SC would increase when either Private cost or/and External Cost increases
- SB would decrease when either private benefit or/and external benefit decreases

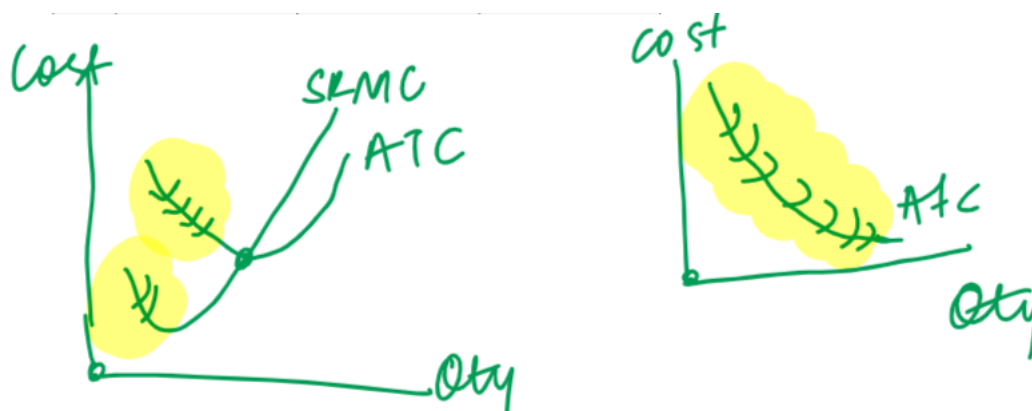
In option A, EB and PB are decreasing, which indicates that SB is falling. Additionally, since PC and EC are unchanged, this demonstrates that SC is the same. With unchanged SC, when SB falls, this represents that Net Social Cost will increase. Hence, key A becomes the right one.

**B** involves an increase in external benefits, which increases Social Benefits and reduces Net Social Cost.

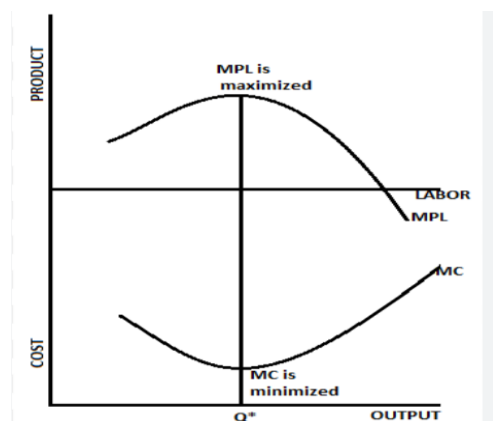
**C** involves an increase in external benefits, which represents rising social benefits. And falling private costs and external costs indicate decreasing social costs. This would reduce Net Social Cost.

**D** shows an increase in private and external benefits, representing a rise in social benefits. Also, a fall in external costs represent a fall in Social cost. This would reduce Net Social Cost.

4-A:



→ As output increases from zero, marginal cost (MC) initially falls due to increasing returns to variable factor (a rise in Marginal product of labor), then rises as diminishing returns set in (a fall in the Marginal Product of Labor). This relationship can also be seen below:



→ Average total cost (ATC) follows a similar U-shape i.e. it decreases first (increasing economies of scale), then increases (decreasing economies of scale).

→ Average fixed cost (AFC), meanwhile, always falls as output increases since fixed costs are spread over more units.

5-D:

Wheat farming is closest to perfect competition: there are many small producers, the product is almost homogeneous (if not completely), and no single farmer can influence market price. Firms are price takers. Therefore, they are least able to alter price.

**Why others are incorrect:**

- **A (Air transport):** This industry has fewer firms and significant pricing power (e.g. via price discrimination, loyalty programs). Air transport is generally cited as an oligopoly.
- **B (Hairdressing):** Differentiated services allow some control over prices. The market structure is generally monopolistic competition.
- **C (Steel):** Steel industry is generally considered an oligopoly. This means that a small number of large firms dominate the market, rather than many competing companies. These firms have a significant influence on market prices and output.

6-C:

- 6 A firm has the choice between five levels of output. The table shows the total cost and total revenue of producing at each output level. The firm could sell whatever output it produces.

|  | output<br>units | total cost<br>\$ | total revenue<br>\$ | profit (TR-TC) |
|--|-----------------|------------------|---------------------|----------------|
|  | 1000            | 8000             | 10000               | 2000           |
|  | 2000            | 12000            | 18000               | 6000           |
|  | 3000            | 19000            | 24000               | 5000           |
|  | 4000            | 23000            | 28000               | 5000           |
|  | 5000            | 25000            | 25000               | 0              |

Handwritten notes:   
- **ATC** (Average Total Cost) is written above the table.   
- **ATC min** is written next to the ATC column, with a bracket indicating the minimum point at 4000 units.   
- **profit (TR-TC)** is written above the profit column.   
- **Total profit max** is written in red, with an arrow pointing to the 6000 profit at 2000 units.   
- **Total revenue max** is written in red, with an arrow pointing to the 28000 revenue at 4000 units.   
- **Sales max** is written in green, with an arrow pointing to the 4000 units output level.

The firm decides to produce 4000 units.

What is the firm's aim?

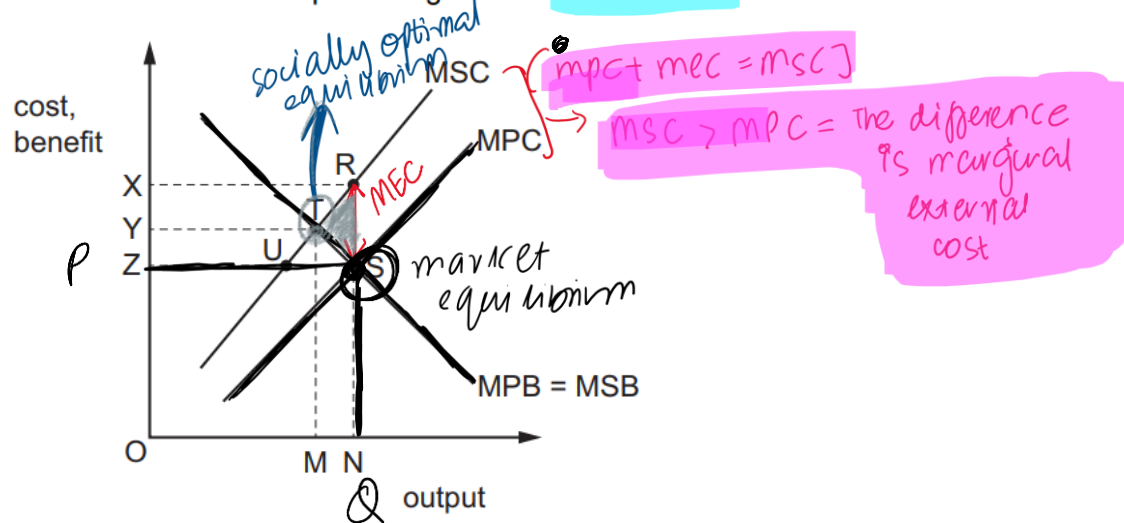
- A to maximise profit
- B to maximise sales
- ☒ C to maximise revenue
- D to minimise average costs

The firm appears to maximize Total Revenue, as at the output level of 4000, it is generating the highest level of TR of 28000.

7-A: *socially optimal equilibrium 'T' where  $MSC = MSB$*

*free market equilibrium 'S' where  $MPC = MPB$*

The diagram shows the costs and benefits of producing steel in a free market.



The area representing deadweight loss (DWL) in externality diagrams is the triangle between marginal social cost (MSC) and marginal social benefit (MSB), over the quantity difference between socially optimal and market output. **In this diagram, RTS is that triangle — showing lost welfare due to overproduction where  $MSC > MSB$ .**

8-C:

Small firms often survive by operating in niche markets or benefiting from lack of competition. If barriers to entry are absent, new firms can easily enter the market, increasing competition and putting downward pressure on prices and profits. This reduces small firm survival chances, thereby making key C correct.

- **A:** A consumer preference for unique, non-standard products supports small businesses that offer differentiation.
- **B:** Lower interest rates make borrowing cheaper, benefiting small firms seeking capital.
- **D:** Decreasing returns to scale mostly affect larger firms, not small ones.

9-A:

A contestable market is one where a few firms exist, but there are low barriers to entry. This creates potential competition even if actual firm numbers are low. The threat of potential competition and fear of 'hit-and-run' entry keeps prices competitive, aligning with the description provided.

### Why others are incorrect:

- **B:** Monopoly involves one dominant firm, not several.
- **C:** Monopolistic competition has many firms (20-25), not a few.
- **D:** Perfect competition has a large number of firms (1000s of them) and perfect knowledge, not “a few.”

10-A:

Backward vertical integration occurs when a firm acquires a supplier. A bakery buying a wheat farm secures its input source, allowing more control over quality, costs, and supply chain efficiency.

### Why others are incorrect:

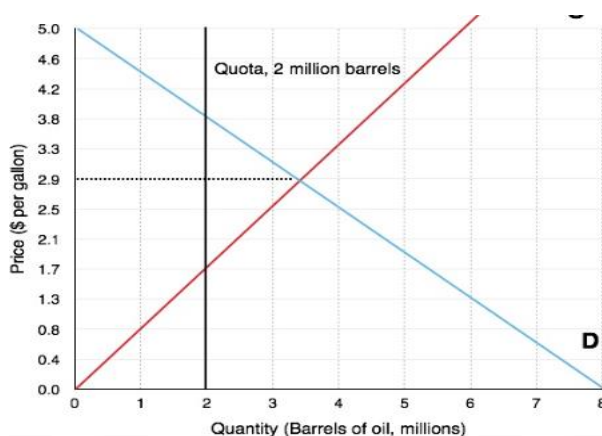
- **B:** This is forward vertical integration — moving closer to the consumer (distribution stage).
- **C:** A vineyard buying an apple farm is an example of diversification.
- **D:** This is horizontal integration — two firms at the same stage of production merging.

11-B:

A regulation banning production is a **non-market-based intervention** because it directly restricts or prohibits activity, rather than working through price signals or incentives. In contrast, market-based solutions influence behavior through costs and benefits (e.g. taxes, tradeable pollution permits).

### Why others are incorrect:

- **A:** A quota limits quantity but still allows price to play a role; market can still operate (as can be seen in the diagram below)



- **C:** A tax internalizes the external cost by making pollution more expensive, aligning private cost with social cost.
- **D:** Tradable pollution permits create a market for emissions, using price incentives to reduce pollution efficiently.

12-D:

**Output per workers employed is the formula of labor productivity.** The table shows that output per worker fell in the shipbuilding industry from 4% to -1% after privatization. A fall in the figure to negative implies declining productivity. This supports the conclusion that workers became less efficient, making D correct.

**Why others are incorrect:**

- **A:** The data is about productivity, not employment levels — we cannot infer employment changes from this.
- **B:** While the output per workers remained constant at 6% in telecommunications, this tells us nothing about the number of workers.
- **C:** Highest productivity growth occurred in electricity supply (from 7% to 9%), but this does not automatically imply greatest absolute output.

13-C:

A subsidized lunch adds to **non-monetary benefits** or **fringe benefits**. The term "net advantage" includes both the wage and the non-monetary aspects of a job. So, introducing subsidised lunches increases the net advantage without altering the wage rate, making C correct.

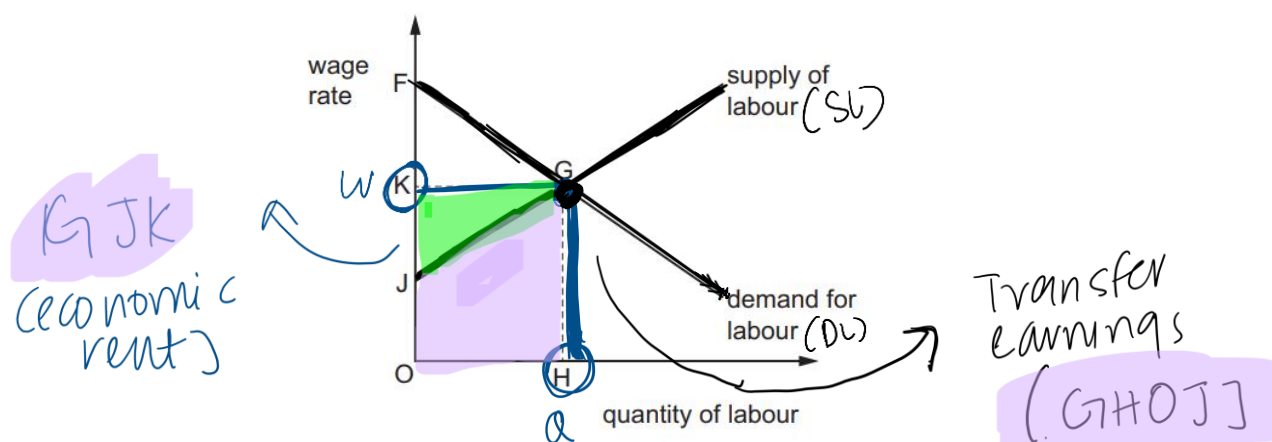
**Why others are incorrect:**

- **A:** Reducing working hours means lower total pay (as workers are paid on an hourly basis, i.e. on the basis of the number of hours worked), likely reducing net advantage.
- **B:** A shift of the economy to the service sector may cause existing workers currently employed in the manufacturing sector to face layoffs.
- **D:** While more jobs might reduce unemployment, it doesn't increase the net advantage for existing workers.

14-C:

A profit-maximising firm will employ labour up to the point where **marginal revenue product (MRP)** equals **marginal cost (MC)** of labour. This is standard theory from the marginal productivity of labour model — hiring additional labour only makes sense as long as it adds more to revenue than it costs.

15-D:



Which areas represent the economic rent and transfer earnings of the farm workers?

|   | economic rent | transfer earnings |
|---|---------------|-------------------|
| A | GFK           | GHOJ              |
| B | GHOJ          | GJK               |
| C | GJK           | GJF               |
| D | GJK           | GHOJ              |

Economic rent is the portion of income earned above the minimum required to keep a worker in their current job, while transfer earnings are the minimum required. On the diagram, area GJK represents economic rent — the amount above the lowest wage required. Area GH OJ shows total transfer earnings, making D the correct combination.

16-A:

Commercial banks do **not** help firms raise finance in terms of issuing equity or underwriting large bond issues — that's the role of **investment banks**, thus making key A the correct option.

On the contrary, commercial banks handle everyday banking: deposits, loans, and payments.

**Why others are incorrect:**

- **B:** Banks do hold deposits for firms — this is core to commercial banking. Commercial banks do hold deposits for firms, along with individuals. These deposits can be in various forms like savings, current, and fixed deposits. These commercial banks then act as financial intermediaries, accepting deposits from firms and individuals and then lending these funds out to borrowers. This allows businesses to manage their finances, make transactions, and access credit.

- **C:** Lending to households is a primary function (e.g. mortgages, personal loans).
- **D:** Savings accounts are fundamental products offered by commercial banks.

17-C:

If the unemployment rate is 4% and employment is 76.6%, then **economically active** population = employed + unemployed = 76.6% + 4% = 80.6%. **This makes key C correct as 80.6% of the age group is economically active.**

Therefore, **economically inactive** = 100% – 80.6% = **19.4%**. So, 23.4% is incorrect. This makes key B incorrect.

**Why others are incorrect:**

- **A:** No data on illegal work is given.
- **D:** The data is internally consistent; there's no evidence it is inaccurate.

18-B:

Unemployment benefits **do not “force” firms to raise wages**. That decision is determined by the labour market, bargaining power, and other factors. Benefits may influence workers' decisions, but **firms are not compelled** to raise wages because benefits exist. **The key word here is “force”**

**Why others are incorrect:**

→ **A:** It's true that benefits can ease the transition for laid-off workers, enabling firms to make workforce adjustments. **The examiner is likely assessing whether you understand that unemployment benefits aren't just for helping workers, but also play a structural role in how flexible or rigid a labor market is. They're part of a system that allows firms to adjust labor needs without causing excessive social harm.**

- **C:** This is a classic argument — benefits may reduce the urgency for job searching, leading to voluntary unemployment.
- **D:** Benefits increase government spending, and if persistent, may lead to higher taxes to balance the budget.

19-A: The formula of multiplier is:

$$\frac{1}{1 - mpc} \quad \text{or} \quad \frac{1}{mpt + mps + mpm}$$

Increasing the **marginal propensity to consume (MPC)** reduces leakages (because an increase in consumption implies that less is being saved now) and boosts the multiplier. A decrease in the **marginal rate of taxation (MRT)** also reduces leakages. Hence, both changes in A increase the multiplier. This is why, key A, becomes correct.

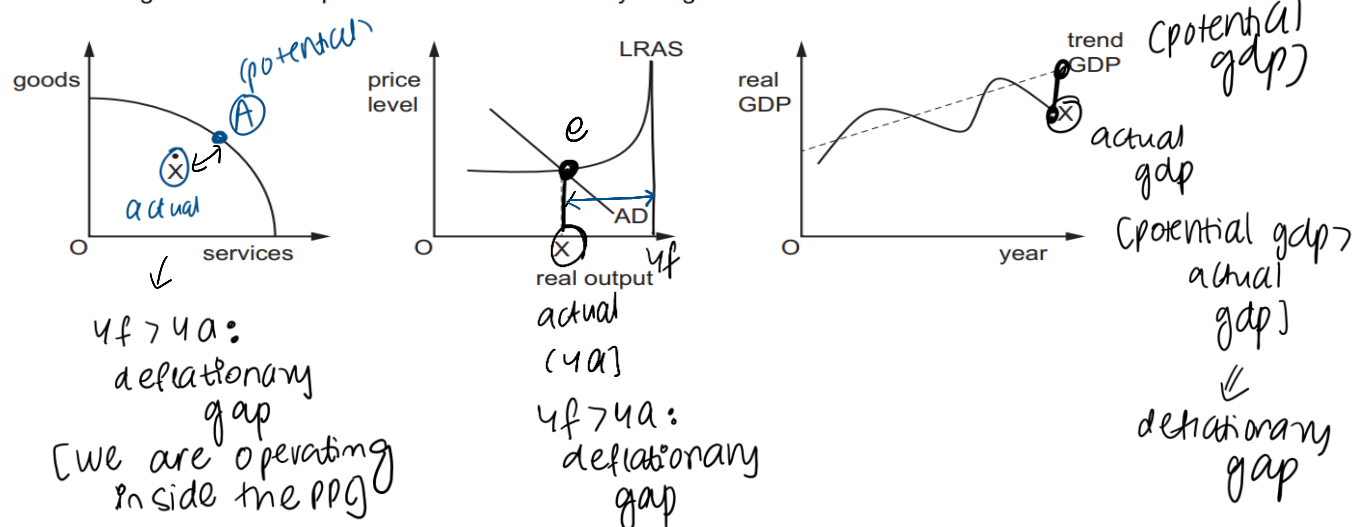
B) A higher MPM increases leakages, lowering the multiplier. A fall in MRT reduces leakages, increasing the multiplier. The overall effect may be nullified.

C) A higher MPS increases leakages, lowering the multiplier. A fall in MPM reduces leakages, increasing the multiplier. The overall effect may be nullified.

D) A higher MRT or MPT increases leakages, lowering the multiplier. A fall in MPS reduces leakages, increasing the multiplier. The overall effect may be nullified.

20-C:

20 The diagrams show the performance of an economy using different measures.



All the diagrams above indicate a **deflationary gap**, where aggregate demand is insufficient to maintain full employment and potential output. The correct policy response is demand-side stimulation. This makes key C correct.

- A) Prices will not increase significantly as the economy continues to grow because there is surplus/unemployment of resources as we are operating far below the potential.
- B) Supply side policies would encourage an increase in the potential of the economy. With unchanged Aggregate demand, an increase in Long-Run Aggregate Supply would increase unemployment further.
- D) There are surpluses of skilled labor in the economy.

21-C:

A liquidity trap is a situation in which **interest rates are very low (near zero)** and monetary policy becomes ineffective. People hoard cash rather than spend or invest.

In this condition, even with low interest, borrowing doesn't increase and neither does investment rise, which means that economic growth remains low. Consequently, inflation rates remain low as well.

**Option C — low growth, low inflation, low interest — accurately reflects this environment.**

22-A:

**To escape a recession, aggregate demand must be increased.** Expansionary fiscal policy (like higher government spending or lower taxes), expansionary monetary policy (lower interest rates), and supply-side measures (long-run productivity improvements) form the strongest combination to stimulate both short- and long-term economic growth.

**Note: Supply-side policies alone won't address Aggregate-demand deficiency.**

Why others are incorrect:

- B: Tight monetary policy would reduce AD and offset any fiscal stimulus arising as a result of procuring expansionary fiscal policies.
- C: Both tight policies would worsen the recession.
- D: Supply-side policies alone won't address Aggregate-demand deficiency.

23-D:

Inflation raises domestic prices, making exports **less competitive** and **imports more attractive**. This typically leads to **increased import expenditure**. Hence, D is the most likely result.

**Why others are incorrect:**

- **A:** It is explained above that inflation would bring about a decrease in export revenues, while simultaneously causing a rise in import expenditures. This represents a worsening of the Balance of Trade. As a result of this, the current account is likely to **worsen**, not improve.
- **B:** Inflation **reduces** price competitiveness as there is a persistent and sustained increase in the general price level of the economy.
- **C:** Export revenues are more likely to fall as foreign buyers face higher prices.

24-D:

To address **deflation**, policy must increase aggregate demand. Reducing interest rates (expansionary monetary policy) and increasing government spending or reducing taxes (expansionary fiscal policy) are appropriate. Thus, option D is correct as it combines both.

**Why others are incorrect:**

- **A:** Raising interest rates would dampen Aggregate demand — the opposite of what's needed.
- **B:** Expansionary fiscal policy could be helpful, but raising interest rates offsets it.
- **C:** Contractionary fiscal policy reduces Aggregate demand, even if interest rates fall.

25-A: *Depreciation, performance in the current account etc are all factors that can potentially affect BOT and then subsequently increase Aggregate demand and a rise in GDP (output) leads to a fall in unemployment. But what primarily drives employment is "GDP" (total output of the economy). Hence, GDP growth is the most direct explanation for a fall in unemployment.*

- A) Argentina's GDP grew by 17.9%, slightly more than Chile's 17.2%. Since both aimed to reduce unemployment by boosting aggregate demand, Argentina's stronger growth likely led to more job creation. A rise in GDP reflects increased output, which is directly linked to a decrease in unemployment, **making this the most direct explanation for Argentina's greater fall in the unemployment rate.**
- B) It is true that Chile's performance on the current account is worse (it has a deficit), while Argentina's performance on the current account was better (it has a surplus). But again, BOT is one component of current account (**Therefore, we can't say that there is a direct link between performance in the current account and unemployment**). All in all, it can't be deduced that there a better performance of Argentina in the current account is the plausible reason why Argentina experienced a greater increase in unemployment.
- C) It's true both countries had a **currency depreciation** against the dollar (Argentina: -18.9%, Chile: -13.3%). But even if Argentina's exchange rate depreciated by a greater percentage, it is not guaranteed that "depreciation" necessarily boosted international trade, and brought about an increase in Aggregate demand, and boosted unemployment. An improvement in BOT position depends on PED of imports and exports. **Depreciation does not have a direct link with unemployment.**

- D) This option introduces wages and inflation, but there's no data given about inflation or wages in the table. Therefore, we cannot assume something not in the data.

26-B:

An **expenditure-switching policy** aims to reduce imports and increase demand for domestic goods by **altering relative prices**, usually through tariffs or devaluation. Encouraging consumers to switch from imports to local goods fits this definition.

**Why others are incorrect:**

- **A:** Expenditure-reducing policies are aimed to dampen overall Aggregate demand of the economy. Contractionary demand-side policies are mainly expenditure-reducing policies.
- **C:** The law of comparative advantage states that individuals, firms, or countries should specialize in producing goods and services where they have a lower opportunity cost than others, and then trade to obtain other goods and services.
- **D:** The Marshall-Lerner condition states that a currency devaluation or depreciation will improve a country's trade balance (the difference between exports and imports) if the sum of the price elasticities of demand for exports and imports is greater than one.

27-A:

he **Marshall-Lerner condition** states that a currency devaluation improves the current account **only if** the sum of the price elasticities of demand for exports and imports is **greater than 1**. That ensures that export volumes rise and import volumes fall enough to improve the balance.

**Why others are incorrect:**

- **B:** If the sum of the price elasticities of demand of imports and exports are less than 1, devaluation will worsen the deficit.
- **C:** Income elasticity relates to income effects, not price effects — irrelevant here.
- **D:** Tariffs imposed by **other countries** in retaliation would hurt exports, not help.

28-C:

When an MNC invests in a developing country, it brings **financial inflows** (FDI), improving the **financial account**. However, if the MNC **imports** its machinery and materials, this increases **import expenditure**, worsening the **current account**.

**Why others are incorrect:**

- **A:** The current account worsens, not improves.

- **B:** Financial account improves, not worsens — FDI is a capital inflow.
- **D:** Worsening of both accounts is inaccurate — the financial account benefits from FDI inflow.

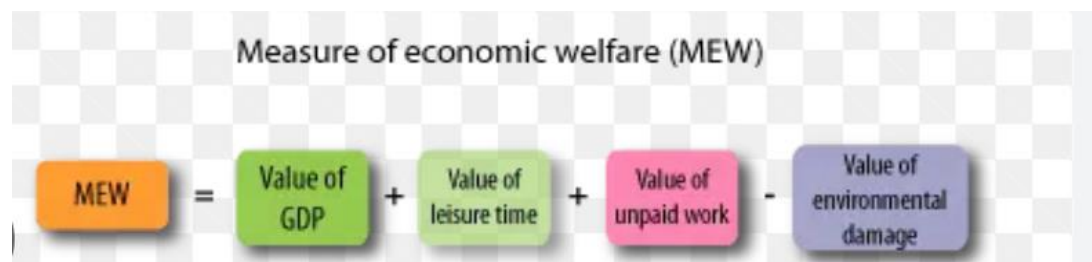
29-A:

The **infant mortality rate** is highest in China (28 per 1000), significantly higher than Hong Kong (8) or Singapore (5). Therefore, it is accurate to conclude that **more children died in infancy** in China that year. This makes option A correct.

**Why others are incorrect:**

- **B:** We need data pertaining to life-expectancy to conclude this.
- **C:** The proportion of population under 16 (**in percentage-terms**) doesn't imply that Singapore has the largest over-16 population in absolute terms.
- **D:** Based on the given data, we can't predict if the population of Singapore was expected to fall in the next five years.

30-D:



Measure of Economic Welfare adjusts the value of GDP to account for factors like leisure time, unpaid work, and environmental impacts, which GDP doesn't account for so that a comprehensive picture of societal well-being can be seen than GDP alone.

The **Measure of Economic Welfare (MEW)** adjusts GDP for environmental degradation and other negative externalities. In this scenario, output per head rose, but pollution worsened — meaning **well-being declined**, even if GDP increased. MEW would reflect this by **falling**. This makes key D correct.

**Why others are incorrect:**

- **A and B:** These purely reflect economic output and would show **increases**, not declines.
- **C:** The HDI focuses on health, education, and income but does not directly measure environmental harm.

